

Bug Fix Impact Analysis

METIMMOX-1 Economic Evaluation

Ben Geisler

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1 Snapshot Inventory

Table 1: Available single-model snapshots

filename	issue	status	commit	mtime
snapshot_145_baseline_ce8ec08.rds	145	baseline	ce8ec08	2026-07-19 21:53:36
snapshot_145_fixed_ea4431f.rds	145	fixed	ea4431f	2026-07-19 21:53:36
snapshot_146_baseline_a3befd2.rds	146	baseline	a3befd2	2026-07-22 21:38:26
snapshot_146_fixed_7f76f9d.rds	146	fixed	7f76f9d	2026-07-22 21:38:26
snapshot_147_baseline_7f76f9d.rds	147	baseline	7f76f9d	2026-07-22 21:38:26
snapshot_147_fixed_d1f203f.rds	147	fixed	d1f203f	2026-07-22 21:38:26
snapshot_149_baseline_8f21e24.rds	149	baseline	8f21e24	2026-09-03 12:39:01
snapshot_149_fixed_4140474.rds	149	fixed	4140474	2026-09-04 00:17:16
snapshot_151_baseline_89d7a28.rds	151	baseline	89d7a28	2026-09-04 01:31:34
snapshot_151_fixed_70d672d.rds	151	fixed	70d672d	2026-09-04 03:06:11
snapshot_152_baseline_e27b60f.rds	152	baseline	e27b60f	2026-09-04 17:45:14
snapshot_152_fixed_9cb460b.rds	152	fixed	9cb460b	2026-09-04 19:17:51
snapshot_153_baseline_66a995c.rds	153	baseline	66a995c	2026-09-05 00:20:04
snapshot_153_fixed_3b291ff.rds	153	fixed	3b291ff	2026-09-05 00:47:49

2 Impact Comparisons

Table 2: Impact comparisons included in this report

Comparison	Before	After
Issue 145: baseline vs fixed	snapshot_145_baseline_ce8ec08.rds	snapshot_145_fixed_ea4431f.rds
Issue 146: baseline vs fixed	snapshot_146_baseline_a3befd2.rds	snapshot_146_fixed_7f76f9d.rds
Issue 147: baseline vs fixed	snapshot_147_baseline_7f76f9d.rds	snapshot_147_fixed_d1f203f.rds
Issue 149: baseline vs fixed	snapshot_149_baseline_8f21e24.rds	snapshot_149_fixed_4140474.rds
Issue 151: baseline vs fixed	snapshot_151_baseline_89d7a28.rds	snapshot_151_fixed_70d672d.rds
Issue 152: baseline vs fixed	snapshot_152_baseline_e27b60f.rds	snapshot_152_fixed_9cb460b.rds
Issue 153: baseline vs fixed	snapshot_153_baseline_66a995c.rds	snapshot_153_fixed_3b291ff.rds
Cumulative: issue 145 baseline vs issue 153 fixed	snapshot_145_baseline_ce8ec08.rds	snapshot_153_fixed_3b291ff.rds

Each comparison below reports the change in base case costs and QALYs, and in net monetary benefit at the willingness-to-pay threshold, from the *before* snapshot to the *after* snapshot. Where both snapshots carry the EVPPI cache (snapshots taken from issue #152 onwards), the per-patient EVPI and the EVPPI table are compared as well; the *after* column includes the Monte Carlo standard error of the regression estimator.

2.1 Issue 145: baseline vs fixed

Table 3: Base case impact – Issue 145: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 20,867	1.4141	EUR 21,056	1.3498	EUR 189	-0.0644
CRP-guided	EUR 55,026	1.3842	EUR 53,791	1.3886	-EUR 1,235	0.0043
TMB/BRAF-guided	EUR 60,798	1.3530	EUR 63,235	1.3514	EUR 2,438	-0.0017

Table 4: Net monetary benefit impact – Issue 145: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 51,254	EUR 47,783	-EUR 3,471
CRP-guided	EUR 15,570	EUR 17,026	EUR 1,456
TMB/BRAF-guided	EUR 8,207	EUR 5,684	-EUR 2,523

2.2 Issue 146: baseline vs fixed

Table 5: Base case impact – Issue 146: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,056	1.3498	EUR 21,056	1.3498	EUR 0	0.0000
CRP-guided	EUR 53,791	1.3886	EUR 53,791	1.3886	EUR 0	0.0000
TMB/BRAF-guided	EUR 63,235	1.3514	EUR 63,235	1.3514	EUR 0	0.0000

Table 6: Net monetary benefit impact – Issue 146: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 47,783	EUR 47,783	EUR 0
CRP-guided	EUR 17,026	EUR 17,026	EUR 0

TMB/BRAF-guided	EUR 5,684	EUR 5,684	EUR 0
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2.3 Issue 147: baseline vs fixed

Table 7: Base case impact – Issue 147: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,056	1.3498	EUR 21,056	1.3498	EUR 0	0.0000
CRP-guided	EUR 53,791	1.3886	EUR 53,791	1.3886	EUR 0	-0.0000
TMB/BRAF-guided	EUR 63,235	1.3514	EUR 63,235	1.3514	EUR 0	-0.0000

Table 8: Net monetary benefit impact – Issue 147: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 47,783	EUR 47,783	EUR 0
CRP-guided	EUR 17,026	EUR 17,026	EUR 0
TMB/BRAF-guided	EUR 5,684	EUR 5,684	EUR 0

2.4 Issue 149: baseline vs fixed

Table 9: Base case impact – Issue 149: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,056	1.3498	EUR 21,523	1.3714	EUR 468	0.0216
CRP-guided	EUR 53,791	1.3886	EUR 53,928	1.3918	EUR 137	0.0033
TMB/BRAF-guided	EUR 63,235	1.3514	EUR 63,399	1.3609	EUR 164	0.0095

Table 10: Net monetary benefit impact – Issue 149: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 47,783	EUR 48,419	EUR 636
CRP-guided	EUR 17,026	EUR 17,056	EUR 30
TMB/BRAF-guided	EUR 5,684	EUR 6,005	EUR 320

2.5 Issue 151: baseline vs fixed

Table 11: Base case impact – Issue 151: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,523	1.3714	EUR 21,523	1.3714	EUR 0	0.0000
CRP-guided	EUR 53,928	1.3918	EUR 53,928	1.3918	EUR 0	0.0000
TMB/BRAF-guided	EUR 63,399	1.3609	EUR 63,399	1.3609	EUR 0	0.0000

Table 12: Net monetary benefit impact – Issue 151: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 48,419	EUR 48,419	EUR 0
CRP-guided	EUR 17,056	EUR 17,056	EUR 0

TMB/BRAF-guided	EUR 6,005	EUR 6,005	EUR 0
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2.6 Issue 152: baseline vs fixed

Table 13: Base case impact – Issue 152: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,523	1.3714	EUR 21,523	1.3714	EUR 0	0.0000
CRP-guided	EUR 53,928	1.3918	EUR 53,928	1.3918	EUR 0	0.0000
TMB/BRAF-guided	EUR 63,399	1.3609	EUR 63,399	1.3609	EUR 0	0.0000

Table 14: Net monetary benefit impact – Issue 152: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 48,419	EUR 48,419	EUR 0
CRP-guided	EUR 17,056	EUR 17,056	EUR 0
TMB/BRAF-guided	EUR 6,005	EUR 6,005	EUR 0

Per-patient EVPI: EUR 45.20 before, EUR 45.20 after.

Table 15: EVPPI impact – Issue 152: baseline vs fixed

Parameter / group	EVPPi Before	% EVPI Before	EVPPi After	SE After	% EVPI After
All costs (group)	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
Drug costs (group)	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
TMB/BRAF-treatment interaction (group)	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
Other costs (group)	EUR 45.20	100.0%	EUR 0.00	EUR 0.04	0.0%
b_tmb_braf_rx_os	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
b_tmb_braf_rx_pfs	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
c_drug_FLOX	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
c_drug_nivo	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
u_np	EUR 45.20	100.0%	EUR 0.00	EUR 0.00	0.0%
c_test_CT	EUR 43.76	96.8%	EUR 0.00	EUR 0.00	0.0%
c_other_last	EUR 41.57	92.0%	EUR 0.00	EUR 0.00	0.0%
p_tmb_braf	EUR 40.09	88.7%	EUR 0.00	EUR 0.00	0.0%
CRP-treatment interaction (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.74	0.0%
Biomarker prevalence (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Test costs (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Utilities (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
b_crp_rx_os	EUR 0.00	0.0%	EUR 0.10	EUR 1.60	0.2%
b_crp_rx_pfs	EUR 0.00	0.0%	EUR 0.00	EUR 0.12	0.0%
c_other_baseline	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_follow	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_visit	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_CRP	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_NGS	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_blood	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
p_crp	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
u_p	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Biomarker-treatment interaction (group)	–	–	EUR 0.00	EUR 1.61	0.0%

2.7 Issue 153: baseline vs fixed

Table 16: Base case impact – Issue 153: baseline vs fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 21,523	1.3714	EUR 21,523	1.3714	EUR 0	0.0000
CRP-guided	EUR 53,928	1.3918	EUR 53,928	1.3918	EUR 0	0.0000
TMB/BRAF-guided	EUR 63,399	1.3609	EUR 63,399	1.3609	EUR 0	0.0000

Table 17: Net monetary benefit impact – Issue 153: baseline vs fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 48,419	EUR 48,419	EUR 0
CRP-guided	EUR 17,056	EUR 17,056	EUR 0
TMB/BRAF-guided	EUR 6,005	EUR 6,005	EUR 0

Per-patient EVPI: EUR 45.20 before, EUR 45.20 after.

Table 18: EVPPI impact – Issue 153: baseline vs fixed

Parameter / group	EVPPPI Before	% EVPI Before	EVPPPI After	SE After	% EVPI After
b_crp_rx_os	EUR 0.10	0.2%	EUR 0.10	EUR 1.60	0.2%
All costs (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Drug costs (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Biomarker-treatment interaction (group)	EUR 0.00	0.0%	EUR 0.00	EUR 1.61	0.0%
CRP-treatment interaction (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.74	0.0%
TMB/BRAF-treatment interaction (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Other costs (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.04	0.0%
Biomarker prevalence (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Test costs (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
Utilities (group)	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
b_crp_rx_pfs	EUR 0.00	0.0%	EUR 0.00	EUR 0.12	0.0%
b_tmb_braf_rx_os	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
b_tmb_braf_rx_pfs	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_drug_FLOX	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_drug_nivo	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_baseline	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_follow	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_last	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_other_visit	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_CRP	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_CT	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_NGS	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
c_test_blood	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
p_crp	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
p_tmb_braf	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
u_np	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%
u_p	EUR 0.00	0.0%	EUR 0.00	EUR 0.00	0.0%

2.8 Cumulative: issue 145 baseline vs issue 153 fixed

Table 19: Base case impact – Cumulative: issue 145 baseline vs issue 153 fixed

Strategy	Cost Before	QALYs Before	Cost After	QALYs After	Cost Change	QALY Change
Standard of Care	EUR 20,867	1.4141	EUR 21,523	1.3714	EUR 656	-0.0427
CRP-guided	EUR 55,026	1.3842	EUR 53,928	1.3918	-EUR 1,098	0.0076
TMB/BRAF-guided	EUR 60,798	1.3530	EUR 63,399	1.3609	EUR 2,602	0.0078

Table 20: Net monetary benefit impact – Cumulative: issue 145 baseline vs issue 153 fixed

Strategy	NMB Before	NMB After	NMB Change
Standard of Care	EUR 51,254	EUR 48,419	-EUR 2,835
CRP-guided	EUR 15,570	EUR 17,056	EUR 1,486
TMB/BRAF-guided	EUR 8,207	EUR 6,005	-EUR 2,203

3 Interpretation

This report compares baseline and fixed single-model snapshots for each bug-fix issue that has both, plus a cumulative comparison from the earliest baseline to the latest fixed snapshot. Snapshots are produced by the `13_save_snapshot.R` analysis script (in `scripts/R/analysis/`). If older snapshots contain obsolete multi-model components, `load_snapshot()` warns during render and the tables use the top-level single-model results.

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